Quick Start Guide
FLIM Kit
FLK-01-1C1L



1 About This Guide

This guide provides the essential steps required to install and commission the FLIM LABS FLIM Kit.

It does **not** replace the FLIM Kit User Manual, the manuals supplied with the individual FLIM LABS components, or the documentation for any connected third-party equipment.

The complete FLIM Kit User Manual is available at:

https://www.flimlabs.com/products/FLIM-kit

Scan the QR code to access the full manual.

Warning Read the full User Manual, all supplied FLIM LABS component manuals, and the documentation of any connected third-party equipment before installation and operation.

2 Safety Information

This kit integrates a pulsed laser source, single-photon detector, timing electronics and data acquisition hardware.

Safe operation requires compliance with:

- FLIM LABS laser safety instructions
- FLIM LABS SPAD detector manual
- FLIM LABS DAQ manual
- Local laboratory safety procedures

Warning Never look into optical fibers, beam paths or laser outputs.

3 Scope of Delivery

The kit is shipped with the following components:

- FLIM LABS pulsed laser
- FLIM LABS SPAD detector
- FLIM Data Acquisition Card
- Optical fibers
- USB cables and power supplies
- Product documentation

Verify that all components of the ordered configuration are present and undamaged before installation.

4 Hardware Installation

Figure 1 shows the complete system connection diagram.

1. Install all components on stable surfaces with sufficient space for ventilation, operation, and cable routing.
2. Connect the pulsed laser source to the experimental setup using the supplied single-mode optical fiber.
3. Connect the fluorescence collection path to the SPAD detector using the supplied multimode optical fiber.
4. Connect the detector output to the detector input of the *FLIM DAQ*.
5. Connect the *Sync-OUT* port of the laser to the *SYNC-IN* input of the *FLIM DAQ*.
6. Connect the *FLIM DAQ* to the acquisition computer using the supplied USB cable.
7. Connect the required power supplies to all FLIM LABS devices.
8. Verify all optical, electrical, USB, and power connections before applying power.

5 Software Configuration

Download and install the latest version of *SpectroscoPy* from:

https://www.flimlabs.com/products/spectroscopy

After completing the hardware installation:

1. Power on all FLIM LABS devices.
2. Start *SpectroscoPy*.
3. Select *Guided Configuration*.
4. Verify that the *FLIM DAQ* is detected.
5. Detect or select the active detector channel.
6. Select the appropriate timing synchronization source.
7. Save the configuration.

For detailed software instructions, refer to the *SpectroscoPy* software manual.

6 First FLIM Measurement

1. Prepare the sample or optical setup.
2. Start data acquisition in *SpectroscoPy*.
3. Verify that photon events are detected.
4. Acquire a fluorescence decay measurement.
5. Adjust the acquisition parameters as required.
6. Save or export the acquired data.

7 Product Label

FLIM Kit

P/N: FLK-01-1C1L
S/N: STK-XXXX-XXX

MADE IN ITALY - **FLIM LABS S.r.l.**
Via della Farnesina 3 - 00135 Rome
techsupport@flimlabs.com

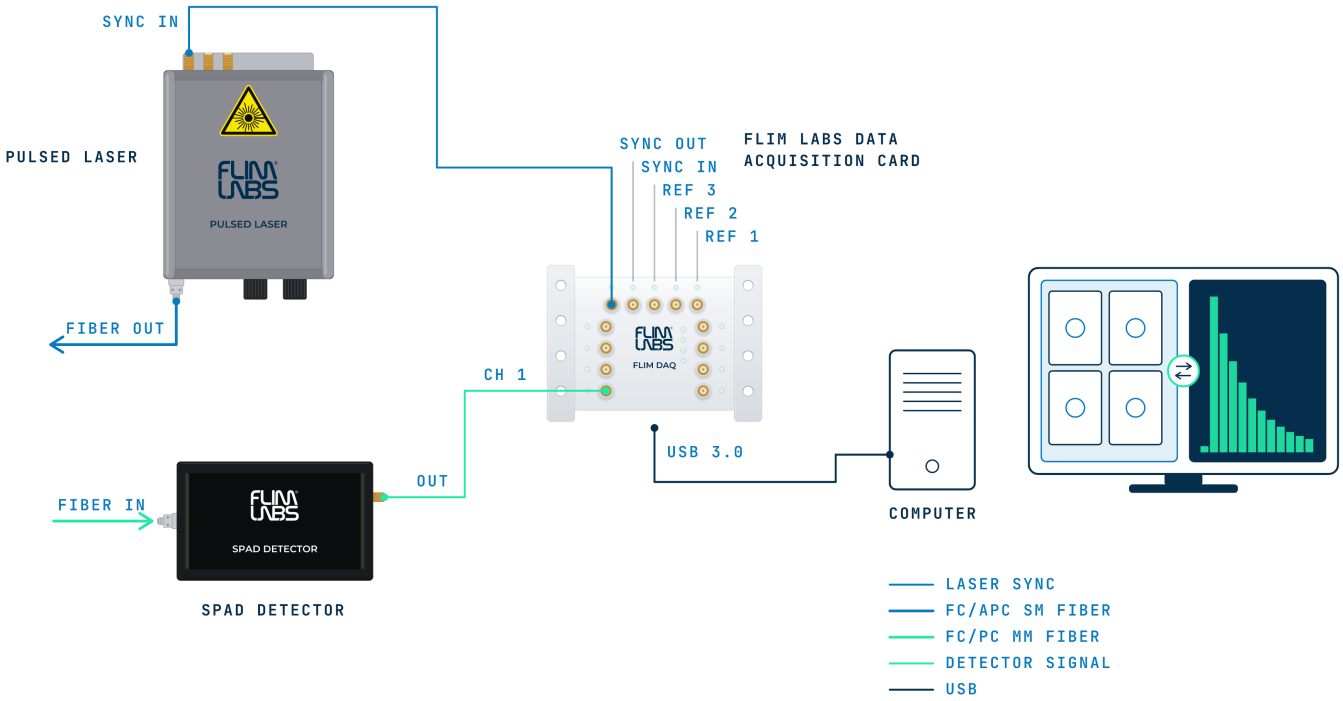


Figure 1: System connection overview.